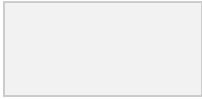


Vision



XQ Secure Workspaces (iOS) —

Executive Summary

AI-Guided Governance

Phased Scope: File Vault + Secure Email + Secure Chat

Core Addition: AI-driven, on-device governance and policy automation (“at the edge”)

Design Principle: Security and compliance happen automatically, without user effort

Platform: iPad & iPhone

[XQ](#) Secure Workspaces is a privacy-first mobile security platform for iOS that combines secure file storage, protected email, private chat and AI-driven governance into a single, frictionless user experience. Built for both consumers and enterprises, the product automatically classifies, protects, and governs sensitive information directly on the device without requiring user training, complex setup, or constant IT oversight.

The platform extends XQ’s Zero Trust Data architecture to mobile environments, enabling organizations to secure documents, communications, and workflows while maintaining consumer-grade simplicity.

XQ Secure Workspaces is an iOS app that combines:

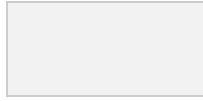
1. **Phase 1 : Secure File Vault**
2. **Phase 2 : Secure Email with protected attachments**
3. **Phase 3: Secure Chat**

Core Value Proposition

XQ uniquely delivers a converged model of:

- **Immediate consumer adoption** with zero setup friction
- **Enterprise-grade governance** with policy-as-code and centralized control
- **Data-centric security enforcement**, where encryption, access control, and auditability are intrinsic properties of the data itself

This eliminates the traditional tradeoff between usability and security, enabling organizations to scale from individual usage to full enterprise deployment within a unified platform.



Overview

The app functions as a secure container: files, emails and chats can be accessed within it, but cannot be exfiltrated from it. XQ holds no customer data at any point, and no content can be moved outside the app to third parties.

Enhanced with **AI-guided experience and governance at the edge**, the app automatically:

- Enable secure collaboration
- Prevents sensitive data leaving
- Enable secure collaboration
- Applies encryption, labeling, and access policies in real time

All enforcement remains aligned with XQ's **Zero Trust Data model**, ensuring compatibility across the XQ platform.

Strategic Outcome + Market Opportunity

The XQ iOS Secure Workspace establishes a new category of mobile-first AI-driven, Zero Trust Data platforms by transforming the mobile device into a governed, encrypted data endpoint. The app enables secure data creation, sharing, and collaboration across consumer and enterprise environments while maintaining consistent enforcement of sovereignty, compliance, and access control policies.

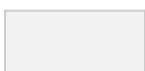
The result is a unified system where data remains protected, portable, and policy-bound at all times, supporting both organic user adoption and mission-critical enterprise deployment at global scale.

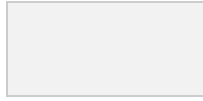
Market Opportunity

The convergence of AI adoption, mobile-first workflows, escalating cyber threats, and tightening global data sovereignty regulations has created a significant market gap between consumer productivity tools and enterprise-grade secure collaboration platforms. Organizations increasingly require solutions that can protect sensitive data beyond traditional network perimeters while enabling seamless user experiences across personal and corporate devices.

XQ is positioned to capture this emerging market by delivering a platform that combines Zero Trust security, AI-native workflows, encrypted collaboration, and policy-enforced data governance in a single mobile-centric architecture. The opportunity spans regulated industries, government and defense, critical infrastructure, global enterprises, and privacy-conscious consumers seeking secure digital workspaces that operate across distributed environments.

As enterprises transition toward AI-enabled operations and decentralized work models, the demand for persistent data protection and sovereign control at the endpoint level is expected to accelerate, creating a scalable platform opportunity for XQ to become foundational infrastructure for secure mobile collaboration and intelligent data exchange.





CORE CAPABILITIES

File & Email and Chat Connectivity

Native connections to SharePoint and network drives. Files are streamed and accessed in-app — never stored on XQ infrastructure.

Secure Viewing & Containment

In-app file viewing with active screenshot blocking and prevention of local download or export. Content cannot be moved outside the app or shared to third-party applications on the device.

AI enabled/supported Onboarding & Account Setup

Streamlined first-run experience with automatic account creation. AI guides users through connecting to SharePoint or a network drive — detecting configuration, resolving common errors, and minimising IT involvement.

Offline Mode

Users can designate files or folders for offline availability. Locally cached copies remain viewable and editable without a connection, with changes syncing back to source when connectivity is restored. Offline cache remains within the secure app container.

Sharing

Sharing via SharePoint (leveraging its existing permissions and link model) and email. All sharing is controlled and auditable — files always resolve from the customer's own infrastructure.

Local File Import

Users can bring in files from iOS local storage (Files app, camera roll, etc.). AI assists by suggesting destination folders, tagging content, and surfacing related files already in connected drives.

AI powered Document scanning

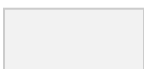
Contextual analysis of file contents runs entirely on-device. No file data leaves the user's device — satisfying enterprise data residency requirements by design.

AI-Driven Policy Engine

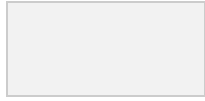
The platform applies configurable policies based on what the on-device AI identifies within files — restricting actions, triggering alerts, or enforcing access rules based on content classification.

Light Editing

Scoped editing for text files, Word documents, and Excel spreadsheets. Changes save directly back to the source location — SharePoint or network drive.



Product Definition



Product Definition

XQ Secure Workspaces delivers effortless security through on-device intelligence.

Users work naturally with files and email while AI continuously:

- Detects sensitive information
- Applies protection policies
- Prevents data leakage
- Guides secure sharing
- Enforces governance automatically

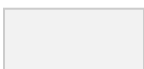
Security becomes ambient, invisible, and continuous rather than manual and disruptive.

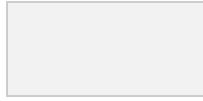
Deployment Model Strategy

The architecture supports three distinct deployment modes:

- **Free Tier (Consumer / Local-First):** On-device encryption using Secure Enclave, no account required, zero infrastructure dependency
- **Enterprise SaaS:** XQ-managed platform with customer-controlled keys, policy engine enforcement, and full auditability

This structure enables bottom-up adoption while supporting top-down enterprise standardization without architectural change.





Deployment Model Strategy

The platform supports three deployment models through a unified architecture.

Free Tier — Consumer / Local-First

A privacy-first standalone deployment designed for individuals and small teams.

Characteristics

- No account required
 - Sharing / connected services
- Local-only operation
- On-device encryption using Apple Secure Enclave
- Zero infrastructure dependency
- AI governance fully local
- No cloud data exposure

Primary Value

Consumer-grade simplicity with automatic protection for sensitive files and communications.

Enterprise SaaS

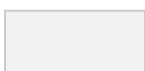
A managed enterprise deployment operated by XQ with customer-controlled governance.

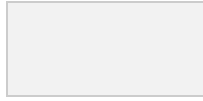
Characteristics

- XQ-managed infrastructure
- Customer-controlled encryption keys
- Enterprise policy engine enforcement
- Full auditability and compliance visibility
- Passive enterprise identity integration
- Centralized administration

Primary Value

Enterprise-scale Zero Trust security with minimal operational friction.





Core Product Components

Secure File Vault

Phase 1

A protected workspace for storing, viewing, editing, and securely sharing files.

Core Capabilities

- Encrypted local file storage
- AI-powered content scanning and classification
- Automatic policy enforcement
- Secure sharing workflows
- Protected attachment handling
- Dynamic access control
- Policy-aware synchronization

Editing Experience

The product intentionally focuses on lightweight productivity rather than full office-suite replacement.

Users can:

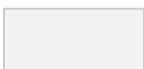
- Open documents
- Make quick edits
- Save changes securely

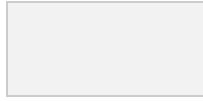
After edits:

- AI automatically rescans content
- Policies update dynamically
- Classification remains current

Supported editing scenarios include:

- Microsoft Word documents
- Excel spreadsheets
- Common mobile productivity workflows





Secure Email

Phase 2

A secure email experience integrated directly into the protected workspace.

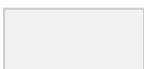
Core Capabilities

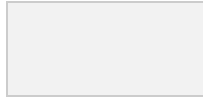
- Email body scanning
- Attachment inspection
- Automatic encryption
- Secure attachment viewing
- AI-assisted risk detection
- Policy-driven message controls

Intelligent Security Behavior

When sensitive information is detected, the system can:

- Recommend secure delivery methods
- Encrypt content automatically
- Restrict forwarding
- Prevent non-compliant sending
- Apply enterprise governance policies





AI-Guided Governance Layer

Phase 1

Overview

AI-guided governance is the foundational intelligence layer of XQ Secure Workspaces.

The platform continuously analyzes content and context directly on-device to automate:

- Classification
- Encryption
- Access control
- Compliance enforcement
- Sharing governance

The user is never expected to manually manage security settings or compliance decisions.

On-Device AI Architecture

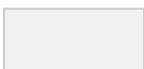
Privacy-First Intelligence

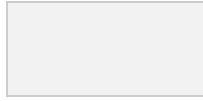
AI models run locally by default.

No raw content leaves the device unless explicitly permitted by enterprise policy.

AI Analysis Capabilities

- Document content scanning
- Metadata inspection
- Email body analysis
- Attachment inspection
- Pattern detection
- Contextual classification
- Policy recommendation





Sensitive Data Detection

The platform identifies:

- Personally Identifiable Information (PII)
- Protected Health Information (PHI)
- Controlled Unclassified Information (CUI)
- Financial records
- Compliance-sensitive content
- NIST

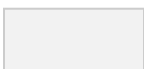
Dynamic Classification

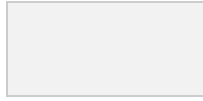
AI continuously classifies content using contextual risk analysis.

Example Classifications

- Public
- Internal
- Confidential
- Restricted
- Custom

Classification automatically drives downstream policy enforcement.





Policy Automation

Consumer Policy Automation

For consumer deployments, governance emphasizes simplicity and passive protection.

Automated Behaviors

- Encrypt all files and emails automatically
- Warn before risky sharing actions
- Recommend secure sharing methods
- Apply local protection silently
- Support secure editing workflows
- Enable file and email revocation

The experience remains lightweight and minimally intrusive.

Enterprise Policy Automation

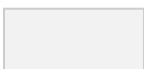
Enterprise deployments introduce centralized governance and dynamic enforcement.

Automated Behaviors

- Restrict external sharing with individuals and groups
- Apply geofencing controls
- Enforce view-only access
- Trigger audit logging
- Apply group-based permissions
- Enforce attribute-based access control (ABAC)
- Prevent policy violations before transmission
- Dynamic policy generation sent to backend

Policies adapt dynamically based on:

- User identity
- Device posture
- Location
- Content sensitivity
- Organizational policy





Real-Time Security Decisioning

Event	Automated Action
File imported	AI scans → classifies → encrypts
Email composed	AI scans → evaluates risk → applies policy->conditionally encrypts
File shared	AI validates recipient, context, and sensitivity, group
Document edited	AI rescans and updates protections
Policy changes	Access rules update dynamically

Security Model

Secure Container Architecture

XQ Secure Workspaces is fundamentally a secure container platform.

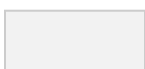
Content enters only through controlled pathways:

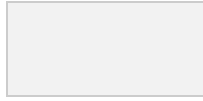
- Microsoft SharePoint
- Local device storage
- Enterprise repositories
- Approved integrations

Data cannot leave the secure workspace except through explicit, auditable sharing workflows.

Data Leakage Prevention

Security protections are foundational rather than optional.





Built-In Controls

- Screenshot blocking where iOS permits
- Sensitive-content watermarking
- Copy/export prevention
- Share-sheet restrictions
- Background blur protection
- Secure attachment handling

AI-Enhanced Protection

High-risk content can trigger stricter runtime protections automatically.

Sharing Experience

Secure Sharing Workflow

The sharing experience is designed to remain simple while enforcing governance automatically.

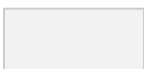
User Flow

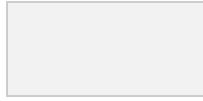
1. User selects “Share Securely”
2. Chooses recipient
3. AI evaluates content sensitivity
4. Appropriate protections apply automatically

AI-Assisted Decisions

The system can:

- **Recommend links vs direct attachments**
- Warn on external recipients
- Detect risky destinations
- Suggest stronger controls
- Restrict sharing to only authorized parties





Enterprise Enforcement

Organizations can automatically apply:

- Expiration policies
- View-only access
- Geofencing
- Access revocation
- Server-side offloading
- Audit tracking
- Labels and classification
- Policy generation

Onboarding Experience

Consumer Onboarding

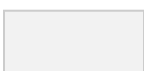
Designed for immediate use with minimal friction.

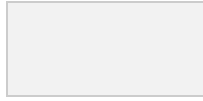
Characteristics

- Instant access
- AI guided onboarding and customization
- No mandatory login
 - Login only required to connect services
 - Optionally User can login to create account
- No required training
- Silent AI governance activation
- Automatic protection on first use

The first imported file automatically triggers:

- Classification
- Encryption
- Governance enforcement





Enterprise Upgrade Path

Consumer deployments can transition seamlessly into enterprise governance.

Upgrade Behaviors

- Existing files are reclassified automatically
- Enterprise policies apply retroactively
- No workflow disruption
- No user retraining required

Interoperability

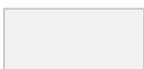
Native Integrations

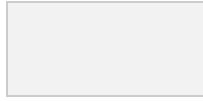
Supported ecosystem integrations include:

- Microsoft SharePoint
- Apple Files (Optional)
- Outlook
- Gmail
- External secure file links

Potential future integrations:

- Google Workspace
- Expanded Microsoft suite
- Additional enterprise repositories





XQ Ecosystem Integration

AI-generated classifications become reusable policy signals across the broader XQ platform.

This enables:

- Consistent policy enforcement
- Shared governance intelligence
- Unified Zero Trust workflows
- Cross-platform interoperability
 - Works with desktop app

User Experience Philosophy

Consumer-Grade Simplicity

The interface is intentionally minimal.

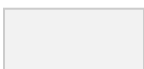
UX Principles

- No policy dashboards for standard users
- No manual classification requirements
- Minimal prompts
- Risk-based intervention only
- Native iOS interaction patterns

User Signals

Security visibility remains lightweight through:

- Small sensitivity labels
- Contextual warnings
- Subtle policy indicators





Enterprise Administration

Administrative Controls

Enterprise administrators can:

- Define classification schemas
- Configure AI sensitivity thresholds
- Override AI decisions
- Monitor policy violations
- Audit data movement
- Configure governance models
- USer access escalation resolution

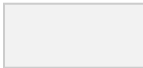
Risk Management

Risk	Mitigation
AI false positives	User overrides + model refinement
Device performance impact	Lightweight on-device models
Privacy concerns	Local processing by default
Workflow disruption	Progressive enforcement and soft warnings

Success Metrics

The platform measures effectiveness through:

- Auto-classification accuracy
- Risky-share prevention rates
- Reduction in data leakage incidents
- User effort per workflow
- Enterprise compliance adherence
- Policy enforcement effectiveness





Final Positioning

XQ Secure Workspaces

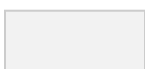
Effortless security powered by AI-driven governance at the edge.

The platform combines:

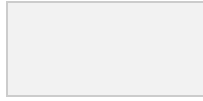
- Secure file storage
- Protected communications
- Real-time AI governance
- Zero Trust enforcement
- Consumer-grade usability

Users work naturally while AI continuously protects data, enforces policy, and prevents risk — without introducing friction or operational complexity.

Should understand if recipient has XQ account for encrypted to encrypted email and file sharing?



Feature Specification



XQ Secure Workspaces —

Core Capability & Product Feature Specification

Product Overview

Secure Workspaces is a secure, AI-governed mobile workspace platform for iOS that combines secure file access, protected communications, lightweight productivity, and autonomous policy enforcement.

The product enables users to work naturally with enterprise files and email while on-device AI continuously:

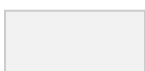
- Detects sensitive information
- Applies governance policies
- Prevents data leakage
- Guides secure sharing
- Enforces Zero Trust controls automatically

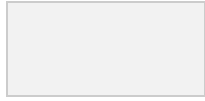
The system is designed around a privacy-first architecture where intelligence operates locally on-device by default and files remain under customer control at all times.

1. Product Objectives

Primary Goals

- Deliver consumer-grade simplicity with enterprise-grade security
- Eliminate manual security workflows wherever possible
- Keep customer data under customer control
- Enable Zero Trust Data governance directly at the edge
- Support both consumer and classified enterprise deployment models through a unified architecture





Key Design Principles

- Security should be ambient and invisible
- AI governance should be passive and automatic
- Users should not manually classify content
- Files should never transit or persist on XQ infrastructure unless explicitly configured
- Policy enforcement should follow the data itself

2. Platform Scope

Supported Platforms

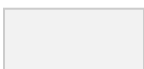
- iPhone
- iPad

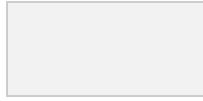
Initial Operating System Support

- iOS (latest two major releases minimum)
 - iOS/iPadOS 17

Future Expansion

- macOS
- Android
- Desktop secure workspace clients





3. Core Product Components

3.1 Secure File Vault

Phase 1

A protected workspace for secure file access, viewing, editing, synchronization, and governed sharing.

Core Responsibilities

- Secure file access
- Encrypted local caching
- AI classification
- Runtime protection
- Controlled sharing
- Secure synchronization
- Policy enforcement

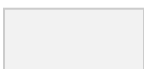
3.2 Secure Email Or Secure Chat

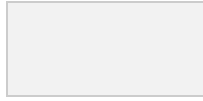
Phase 2

Integrated secure email experience operating within the protected workspace.

Core Responsibilities

- Secure email composition
- Attachment scanning
- AI-driven risk analysis
- Automatic encryption
- Secure attachment rendering
- Governance enforcement
- Group sharing





3.3 AI-Guided Governance Layer

Central intelligence layer responsible for:

- Content analysis
- Classification
- Policy automation
- Risk detection
- Dynamic access enforcement
- Encryption enforcement

All AI processing occurs locally on-device by default.

4. Core Capability Specifications

4.1 File Connectivity

Description

The platform provides native connectivity to enterprise repositories while ensuring files remain under customer ownership and control.

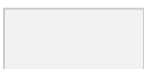
Supported Sources

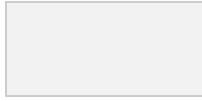
Initial Support

- Microsoft SharePoint
- SMB / network drives
- Apple Files local storage

Future Integrations

- Google Workspace
- OneDrive
- Box
- Dropbox





- Enterprise ECM systems

Functional Requirements

Repository Access

- Authenticate using enterprise credentials
- Browse remote directories
- Stream files directly into the secure workspace
- Maintain source-of-truth at origin repository

Security Requirements

- Files must never persist on XQ-managed infrastructure
- Remote files stream directly to device
- Session tokens stored securely in iOS Keychain/Secure Enclave

Offline Handling

- Explicit user-selected offline caching only
- Cache remains encrypted within app container

Sharing

- Create groups and share files and folders by inviting participants inside and outside your organization

ScreenShot

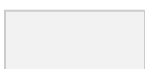
- Prevent in app screen shot or watermark

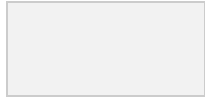
Secure Viewing / Editing

- Docs can be opened and edited in app only

Enclave

- Files may not be available to the OS file system. Architecture dependent.





4.2 Secure Viewing & Containment

Description

All files open within a hardened secure container environment.

Runtime Protections

Mandatory Protections

- In-app secure rendering only
- Disable “Open In” workflows
- Prevent export to third-party applications
- Restrict copy/paste operations
- Block unapproved share sheet destinations
- Blur application during backgrounding
- Watermark sensitive content dynamically

Screenshot Protection

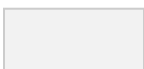
Where iOS APIs permit:

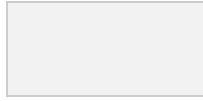
- Screenshot blocking / watermarking
- Screen recording detection
- Runtime sensitivity escalation

Data Containment Rules

Files may only leave the workspace through:

- Approved secure sharing workflows
- Policy-approved email delivery
- Repository-backed sharing links





4.3 AI-Enabled Onboarding & Account Setup

Description

The onboarding flow minimizes user friction through AI-assisted setup and automated configuration.

User Experience Goals

- Zero-training onboarding
- Minimal IT involvement
- Fast first-run activation

Functional Capabilities

Automated Setup

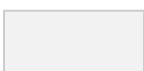
AI assists users with:

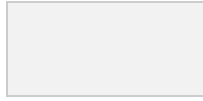
- Onboarding and user training'
- SharePoint connection setup
- Network drive discovery
- Local file import
 - Scan files on device and import them?
- Credential validation
- Common configuration remediation
- UX PReferences

Intelligent Assistance

AI may:

- Detect repository type
- Suggest connection settings
- Resolve authentication issues
- Recommend sync configurations





Enterprise Identity Support

- SSO support
- Passive enterprise identity integration
- Conditional access compatibility
 - Entra
 - Gmail
 - AD
 - Okta
 - Ping

4.4 Offline Mode

Description

Users may explicitly designate files or folders for offline access.

OR set all for default offline access?

Functional Requirements

Offline Availability

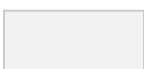
Users can:

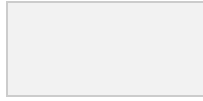
- Mark files/folders offline
- View cached content offline
- Edit supported file types offline

Synchronization

When connectivity returns:

- Local changes sync automatically
- Conflict resolution prompts user only when necessary
- Policy enforcement re-evaluates modified content





Security Requirements

- Offline cache encrypted locally
- Cache inaccessible outside app container
- Enterprise policy can disable offline mode

4.5 Sharing

Description

Sharing workflows maintain governance while remaining user-friendly.

Sharing Methods

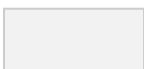
- SharePoint link sharing
- Group peer to peer
- Secure email sharing
- Governed attachment delivery

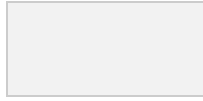
AI-Assisted Sharing Controls

AI Evaluations

Before sharing:

- Analyze content sensitivity
- Validate recipient trust level
- Detect external recipients
- Evaluate policy compliance
- Suggest group recipients





Automated Enforcement

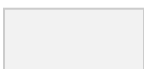
System may:

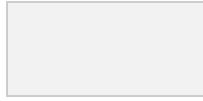
- Suggest policies
- Assist sharing
- Recommend secure links over attachments
- Force encryption
- Restrict forwarding
- Apply expiration policies
- Enforce geofencing
- Block non-compliant sharing

Auditability

All enterprise sharing events must support:

- Audit logging
- User attribution
- Policy decision traceability





4.6 Local File Import

Description

Users may securely import files from iOS-native sources.

Supported Sources

- Apple Files app
- Camera roll
- Local downloads
- Device storage

AI-Assisted Import

AI Behaviors

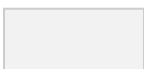
Upon import:

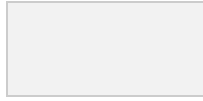
- Scan file contents
- Suggest repository destinations
- Recommend tags/classification
- Surface related files
- Organize based on content

Governance Actions

Imported files automatically:

- Encrypt locally
- Receive classification labels
- Trigger policy enforcement





4.7 AI-Powered Document Scanning

Description

Content analysis executes entirely on-device by default.

Privacy Requirements

- No raw content leaves device unless enterprise policy permits
- No cloud AI dependency required
- All scanning local-first

Supported Analysis

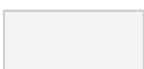
Content Inspection

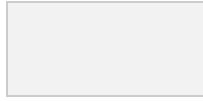
- Text extraction
- Metadata inspection
- Attachment analysis
- Phase 3:
 - OCR for scanned documents/images

Sensitive Data Detection

System detects:

- PII
- PHI
- CUI
- Financial records
- Compliance-sensitive material
- NIST
- Phase 3:
 - CJIS





4.8 AI-Driven Policy Engine

Description

Policies dynamically adapt based on AI classification and runtime context.

Policy Inputs

- Content sensitivity
- User identity
- Device posture
- Geographic location
- Enterprise attributes
- Sharing destination

Automated Actions

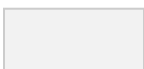
Enforcement Examples

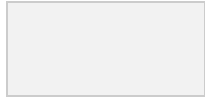
- Restrict external sharing
- Apply view-only mode
- **Disable offline access**
- Trigger alerts
- Enforce ABAC/RBAC rules
- IDP Attributes
- Prevent transmission
- Support group sharing

Dynamic Re-Evaluation

Policies update automatically when:

- Content changes
- User context changes
- Device posture changes
- Enterprise policies update





4.9 Light Editing

Description

The platform supports lightweight productivity workflows without replacing full desktop office suites.

Supported Editing

- Text files
- Microsoft Word documents
- Excel spreadsheets

Functional Requirements

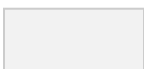
Users can:

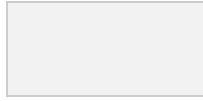
- Open documents
- Make lightweight edits
- Save directly to source repository

Post-Edit Processing

After save:

- AI rescans content
- Classification updates dynamically
- Policies re-evaluate automatically





5. AI Governance Architecture

5.1 On-Device AI Model Strategy

Requirements

- Local inference by default
- Lightweight runtime footprint
- Battery-aware processing
- Offline-capable governance

AI Responsibilities

- Classification
- Risk scoring
- Policy recommendation
- Sensitive data detection
- Sharing risk evaluation

5.2 Dynamic Classification System

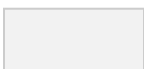
Classification Labels

Examples include:

- Public
- Internal
- Confidential
- Restricted
- *custom

Requirements

- Continuous reclassification
- Context-aware sensitivity evaluation
- Multi-label support





5.3 Real-Time Security Decisioning

Event	Automated Action
File imported	Scan → classify → encrypt
Email composed	Scan → risk evaluate → enforce policy
File shared	Validate recipient/context
Document edited	Rescan + update protections
Policy changed	Recalculate access rules

6. Deployment Models

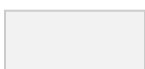
6.1 Free Tier — Consumer / Local-First

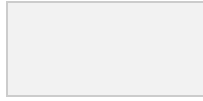
Characteristics

- No account required
- Fully local operation
- Secure Enclave encryption
- No cloud dependency
- Local AI governance

Primary Use Case

Privacy-focused consumers and small teams.





6.2 Enterprise SaaS

Characteristics

- XQ-managed infrastructure
- Customer-controlled encryption keys
- Centralized governance
- Auditability
- Enterprise policy enforcement

Primary Use Case

Enterprise Zero Trust deployments.

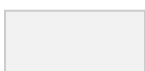
6.3 Phase 3: Self-Hosted / Ghosted Deployment

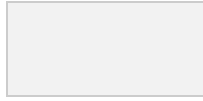
Characteristics

- Fully customer-operated
- No XQ visibility into runtime operations
- Air-gapped capable
- Classified-environment compatible

Primary Use Case

Government, defense, regulated environments.





7. Security Architecture

7.1 Secure Container Model

Data Entry Points

Approved pathways only:

- SharePoint
- Enterprise repositories
- Local import
- Approved integrations

Data Egress Controls

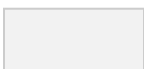
Data may only leave through:

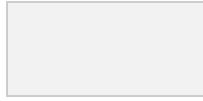
- Governed sharing
- Policy-approved email
- Authorized repository synchronization

7.2 Data Loss Prevention (DLP)

Built-In Protections

- Screenshot mitigation
- Watermarking
- Export restrictions
- Copy/paste restrictions
- Share sheet filtering
- Background obfuscation





AI-Enhanced Runtime Controls

High-risk content may dynamically trigger:

- Stronger viewing restrictions
- Additional authentication
- Share prevention

8. Enterprise Administration

Administrative Capabilities

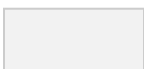
Administrators can:

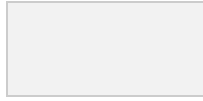
- Define classifications
- Configure sensitivity thresholds
- Override AI decisions
- Configure governance rules
- Audit activity
- Manage policy models
- Security Groups mapped to policies and labels

Governance Models

Support for:

- RBAC
- ABAC
- Conditional access
- Geofencing
- Dynamic trust evaluation





9. User Experience Philosophy

UX Principles

- Minimal friction
- Native iOS interaction patterns
- Silent governance
- Contextual warnings only when needed
- Agentic first

User-Facing Indicators

- Sensitivity labels
- Lightweight policy indicators
- Risk-based prompts

10. Interoperability

Native Integrations

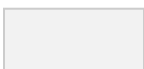
- Microsoft SharePoint
- Apple Files
- Outlook

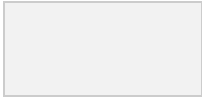
Ecosystem Integration

AI-generated classifications become reusable policy signals across the broader XQ ecosystem.

Enables:

- Unified governance
- Cross-platform enforcement
- Shared Zero Trust intelligence





11. Risk Management

Risk	Mitigation
AI false positives	User override + model refinement
Performance impact	Lightweight local models
Privacy concerns	Local-first processing
Workflow disruption	Progressive enforcement

12. Success Metrics

Product Metrics

- Auto-classification accuracy
- Risky-share prevention rate
- Reduction in leakage incidents
- User friction reduction
- Policy enforcement effectiveness
- Compliance adherence
- Avoiding AI hallucinations

Application Specification

XQ Secure Workspaces — Application Sitemap & Screen Specification

1. Application Structure Overview

The application is organized around five primary navigation domains:

1. Home Workspace
2. Files & Repositories
3. Secure Email
4. Sharing & Activity
5. Settings & Governance

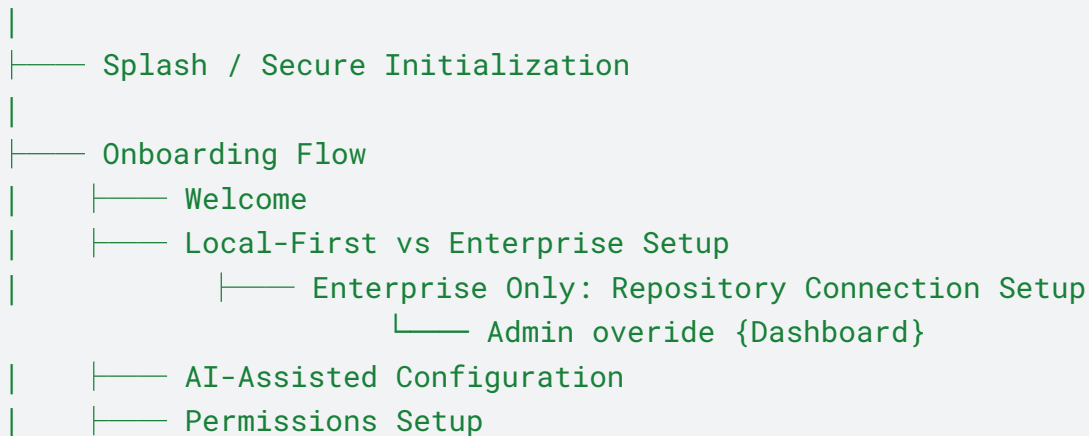
The product prioritizes:

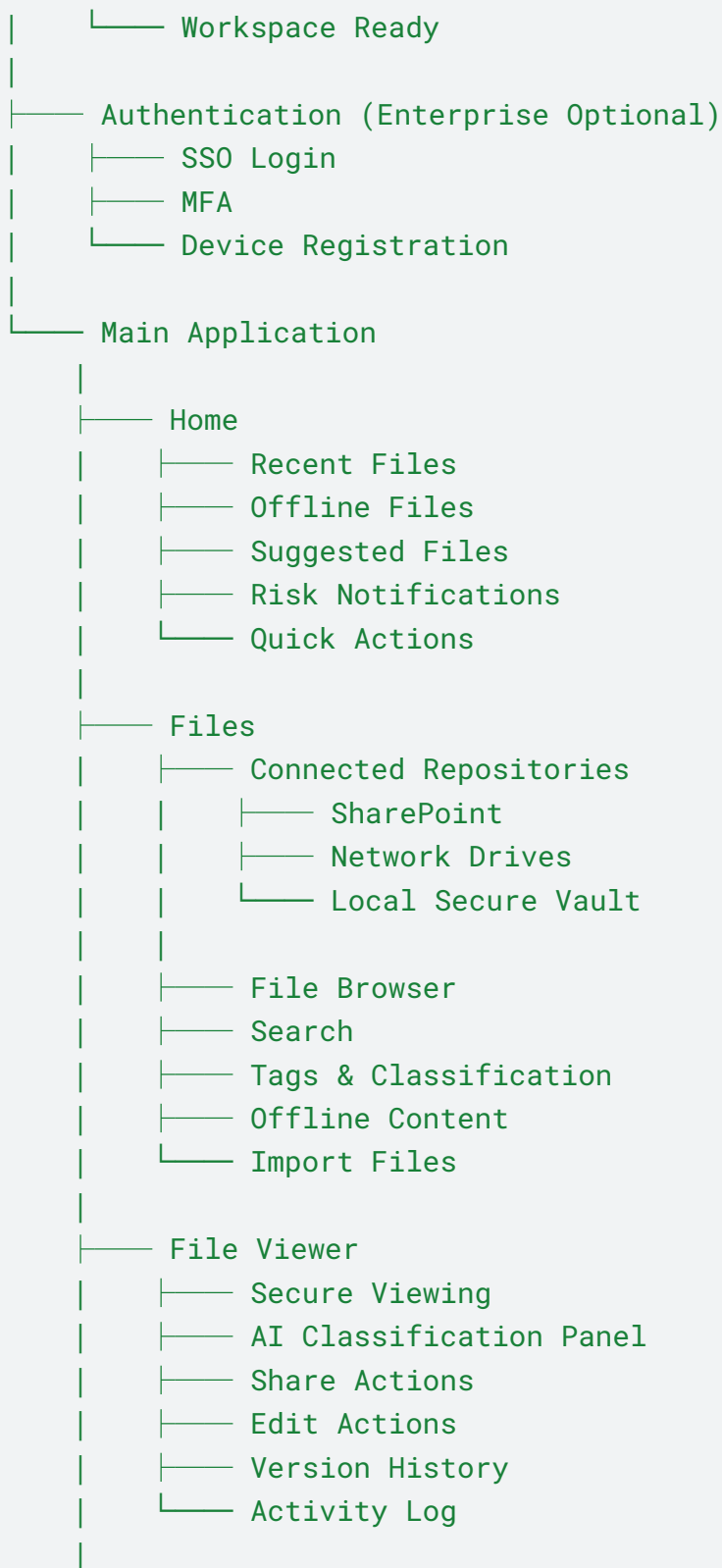
- Minimal navigation complexity
- Contextual actions
- Ambient AI governance
- Native iOS interaction patterns

2. High-Level Sitemap

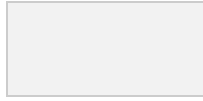
None

Launch





- └─ Secure Email
 - └─ Inbox
 - └─ Compose
 - └─ Attachments
 - └─ Secure Send Controls
 - └─ Message Viewer
- └─ Sharing Center
 - └─ Shared Files
 - └─ Active Links
 - └─ Revocation Controls
 - └─ External Recipients
 - └─ Share Audit Log
- └─ Notifications
 - └─ Risk Alerts
 - └─ Policy Events
 - └─ Sync Events
 - └─ AI Recommendations
- └─ Enterprise Admin (Role Based)
 - └─ Policy Management
 - └─ Classification Models
 - └─ Audit Logs
 - └─ User Access
 - └─ Device Trust
 - └─ Governance Rules
- └─ Settings
 - └─ Account
 - └─ Security
 - └─ Offline Storage
 - └─ AI Governance Preferences
 - └─ Repository Connections
 - └─ Notifications
 - └─ About



3. Screen Specifications

3.1 Splash / Secure Initialization Screen

Purpose

Initial secure app boot and runtime integrity validation.

Functional Components

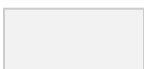
- XQ logo animation
- Secure environment validation
- Device trust checks
- Secure enclave initialization
- Token restoration
- Offline cache verification

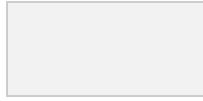
Background Actions

- Validate app integrity
- Restore encrypted session
- Verify enterprise policies
- Initialize local AI models

Failure States

- Jailbreak/root detection
- Integrity failure
- Expired session
- Device non-compliance





3.2 Welcome / Onboarding Screen

Purpose

Introduce the product and establish deployment model.

UI Components

- Intro slides
- “Continue Local-First”
- “Connect Enterprise Workspace”
- Privacy-first messaging

Functional Actions

- Select deployment type
- Begin guided setup
- Trigger AI onboarding assistant

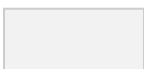
3.3 AI-Assisted Repository Setup Screen

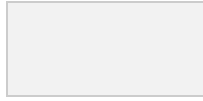
Purpose

Configure SharePoint or network drive access.

UI Components

- Repository type selector
- AI setup assistant chat panel
- Connection diagnostics
- Auto-discovered repositories
- Credential forms





Functional Capabilities

AI Assistance

- Detect SharePoint configuration
- Suggest URLs
- Resolve authentication errors
- Recommend sync scope

Connectivity Features

- OAuth/SSO login
- SMB configuration
- Token validation
- Connectivity testing

3.4 Permissions Setup Screen

Purpose

Request required iOS permissions.

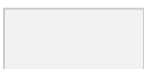
Permissions

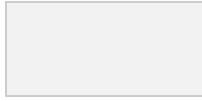
- Face ID
- Notifications
- Files access
- Camera roll import
- Local network access (if needed)

AI Guidance

Explain:

- Why permissions are needed
- Security/privacy implications
- Enterprise policy impacts





3.5 Home Screen

Purpose

Primary productivity dashboard.

Layout Sections

Recent Files

- Recently accessed documents
- Quick resume actions

Suggested Files

AI-generated suggestions based on:

- Recent activity
- Related projects
- Sharing behavior

Offline Files

- Cached content
- Sync status indicators

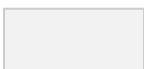
Risk Notifications

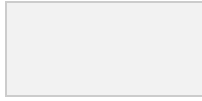
Examples:

- “Sensitive file shared externally”
- “Unclassified document detected”

Quick Actions

- Import file
- Compose secure email
- Share securely





- Scan document

3.6 Repository Browser Screen

Purpose

Unified file access experience.

UI Components

- Repository switcher
- Folder hierarchy
- File list/grid
- Classification badges
- Search bar
- Filter controls

File Indicators

- Sensitivity labels
- Offline availability
- Share status
- Sync state

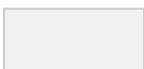
Functional Capabilities

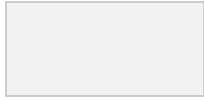
Navigation

- Browse repositories
- Multi-select files
- Search across repositories

AI Features

- Suggested folders
- Related content recommendations
- Smart tagging





3.7 Secure File Viewer Screen

Purpose

Protected in-app document rendering.

Supported Formats

- PDF
- DOCX
- XLSX
- TXT
- Images

UI Components

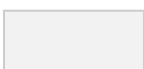
- Document canvas
- Sensitivity label
- Policy indicators
- Watermark overlay
- Share button
- Edit button
- Activity button

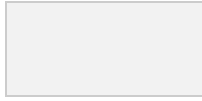
Runtime Protections

- Screenshot blocking
- Screen recording detection
- Background blur
- Copy prevention
- Share restriction enforcement

AI Features

Contextual Intelligence





- Sensitivity analysis
- Risk explanation
- Policy rationale

Dynamic Protection

High-risk content may:

- Disable sharing
- Enforce view-only
- Trigger additional authentication

3.8 Document Editing Screen

Purpose

Lightweight productivity editing.

Supported Actions

Word

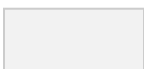
- Text edits
- Formatting
- Comments

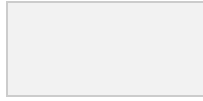
Excel

- Cell editing
- Formula support
- Basic formatting

Functional Capabilities

- Auto-save
- Secure local cache
- Repository synchronization





AI Behaviors

After edits:

- Rescan document
- Reclassify sensitivity
- Reapply governance rules

3.9 Secure Share Workflow Screen

Purpose

Governed secure sharing experience.

User Flow

1. Select recipient
2. AI evaluates risk
3. Policies apply automatically
4. User confirms

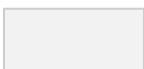
UI Components

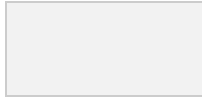
- Recipient selector
- Share method options
- Security settings
- AI risk summary
- Expiration controls

AI Risk Signals

- External recipient warning
- Sensitive content alert
- Compliance risk detection

Enterprise Controls





Admins may enforce:

- View-only access
- Expiration
- Geofencing
- Download prevention

3.10 Local File Import Screen

Purpose

Bring external content into secure workspace.

Import Sources

- Apple Files
- Camera roll
- Scan document
- Downloads

AI Features

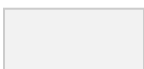
Automatic Suggestions

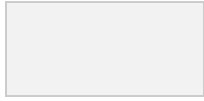
- Recommended destination folder
- Classification predictions
- Related file detection

Background Actions

- Encrypt imported file
- Scan content
- Apply policy protections

3.11 AI Document Scanner Screen





Purpose

Capture and analyze physical documents.

UI Components

- Camera interface
- OCR preview
- Sensitivity detection panel
- Classification preview

AI Features

Detection

- PII
- PHI
- Financial information
- CUI patterns

Actions

- Encrypt immediately
- Suggest secure repository
- Apply governance labels

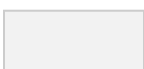
3.12 Secure Email Inbox Screen

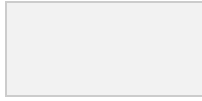
Purpose

Protected enterprise email experience.

UI Components

- Inbox list
- Risk indicators
- Secure attachment badges





- Sensitivity labels

Functional Capabilities

- Open messages securely
- Scan attachments
- Govern forwarding/sharing

AI Behaviors

- Detect risky messages
- Flag sensitive attachments
- Recommend secure response methods

3.13 Compose Secure Email Screen

Purpose

Governed email composition.

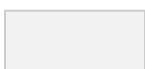
UI Components

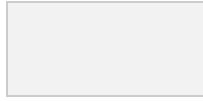
- Recipient fields
- Subject/body editor
- Attachment controls
- AI security recommendations

AI Analysis

Continuously evaluates:

- Message content
- Attachments
- Recipients
- Compliance posture





Automated Enforcement

May:

- Force encryption
- Block send
- Warn user
- Restrict recipients

3.14 Notifications & Security Events Screen

Purpose

Centralized awareness and governance visibility.

Notification Categories

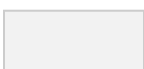
- Sharing risks
- Policy violations
- Sync conflicts
- Access changes
- AI recommendations

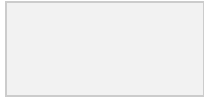
Functional Capabilities

- Review events
- Resolve warnings
- Retry failed syncs
- Escalate incidents

3.15 Sharing Center Screen

Purpose





Manage active shares and external access.

UI Components

- Shared file list
- External recipient list
- Active secure links
- Expiration timers

Functional Capabilities

- Revoke access
- Extend expiration
- View audit activity
- Restrict permissions

3.16 Enterprise Policy Management Screen

Access

Enterprise administrators only.

Purpose

Configure governance and security policies.

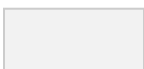
Functional Areas

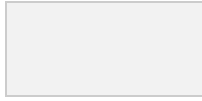
Classification Management

- Define labels
- Configure AI thresholds

Sharing Policies

- External sharing rules





- Geofencing
- Expiration defaults

Runtime Protection

- Screenshot enforcement
- Offline restrictions
- Copy/export controls

AI Governance Controls

- Confidence thresholds
 - Override rules
 - Escalation behavior
-

3.17 Audit & Activity Screen

Purpose

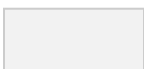
Enterprise audit visibility.

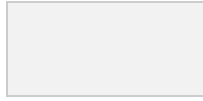
Event Types

- File access
- Sharing events
- Policy violations
- Classification changes
- Device trust events

Features

- Search/filter logs
 - Export audit reports
 - User activity tracing
-





3.18 Settings Screen

Categories

Account

- Identity
- SSO status
- Device registration

Security

- Face ID
- Session timeout
- App lock

Offline Storage

- Cache management
- Offline folder selection

AI Governance

- Consumer-level preferences
- Privacy settings

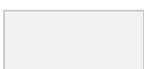
Repository Connections

- Add/remove repositories
- Reauthenticate connections

Notifications

- Alert preferences
- Risk notification levels

4. Navigation Model





Primary Navigation (Bottom Tab Bar)

Tab	Purpose
Home	Productivity dashboard
Files	Repository/file access
Email	Secure communications
Shares	Sharing management
Settings	Configuration

5. AI Interaction Model

Ambient AI

AI should operate silently whenever possible.

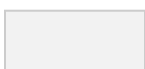
Explicit AI Interactions

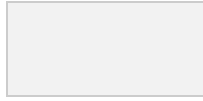
Only surface when:

- Risk detected
- User decision needed
- Policy conflict exists

AI Presentation Style

- Minimal prompts
- Contextual suggestions
- Human-readable reasoning
- Non-technical language





6. Role-Based UX Variants

Standard User

- Minimal controls
- Guided governance
- Passive security

Enterprise Admin

Additional visibility:

- Policy controls
- Audit visibility
- Governance analytics
- AI tuning controls

7. Security UX Requirements

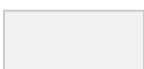
Mandatory Runtime Behaviors

- Blur app on multitask switch
- Require biometric unlock on return
- Restrict screenshots where possible
- Prevent external app handoff

Sensitive Content Escalation

High-risk content may dynamically:

- Increase watermark visibility
- Require reauthentication
- Restrict sharing actions

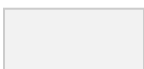




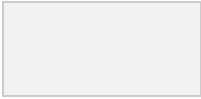
8. Future Screen Expansion Areas

Planned Future Modules

- Teams collaboration
- Secure chat
- AI governance analytics
- Workflow automation
- Cross-device workspace sync
- Secure browser mode
- Enterprise DLP dashboard



Technical considerations

- 
1. Support the Enforcement of policies, groups, labels and IDP attributes
 2. No direct API calls, everything through an interface
 3. Interchangeable AI models
 4. Interface for XQ API that will support multiple versions.
 5. mULTI LANGUAGE SUPPORT
 - a. External LANGUAGE file for all UI/US messages to make translation easier
 - b. ADAPTABLE ui TEXT FIELD AND BUTTONS
 6. Resource control
 7. AUtom update / force update
 8. INTERface for all INtegrations - MSFT, GOOGLE
 9. Every UX element has an animation for entry and exit and click.
 10. “Fuzzy logic for POlICY implementation”
 11. Encryped files
 12. Cross platform considerations
 13. Gmail is group ID
 14. INVITE FOLKS AND IT BECOMES A SHARED FOLDER

XQ Secure Workspaces AI

Technical Architecture & Mobile Application Specification

iOS / iPadOS Enterprise Zero Trust Data Platform

1. System Overview

Product Objective

XQ Secure Workspaces is an AI-native Zero Trust mobile platform for secure file access, secure communications, governed collaboration, and persistent data protection across enterprise and sovereign environments.

The application provides:

- AI driven UX
- Secure repository access
- AI-driven governance

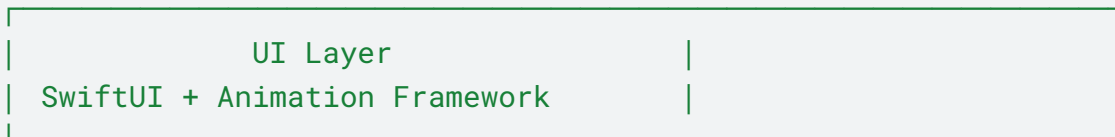
- Persistent encryption
- Dynamic policy enforcement
- Cross-platform extensibility
- Modular integration architecture
- Offline-first protected workspace capabilities

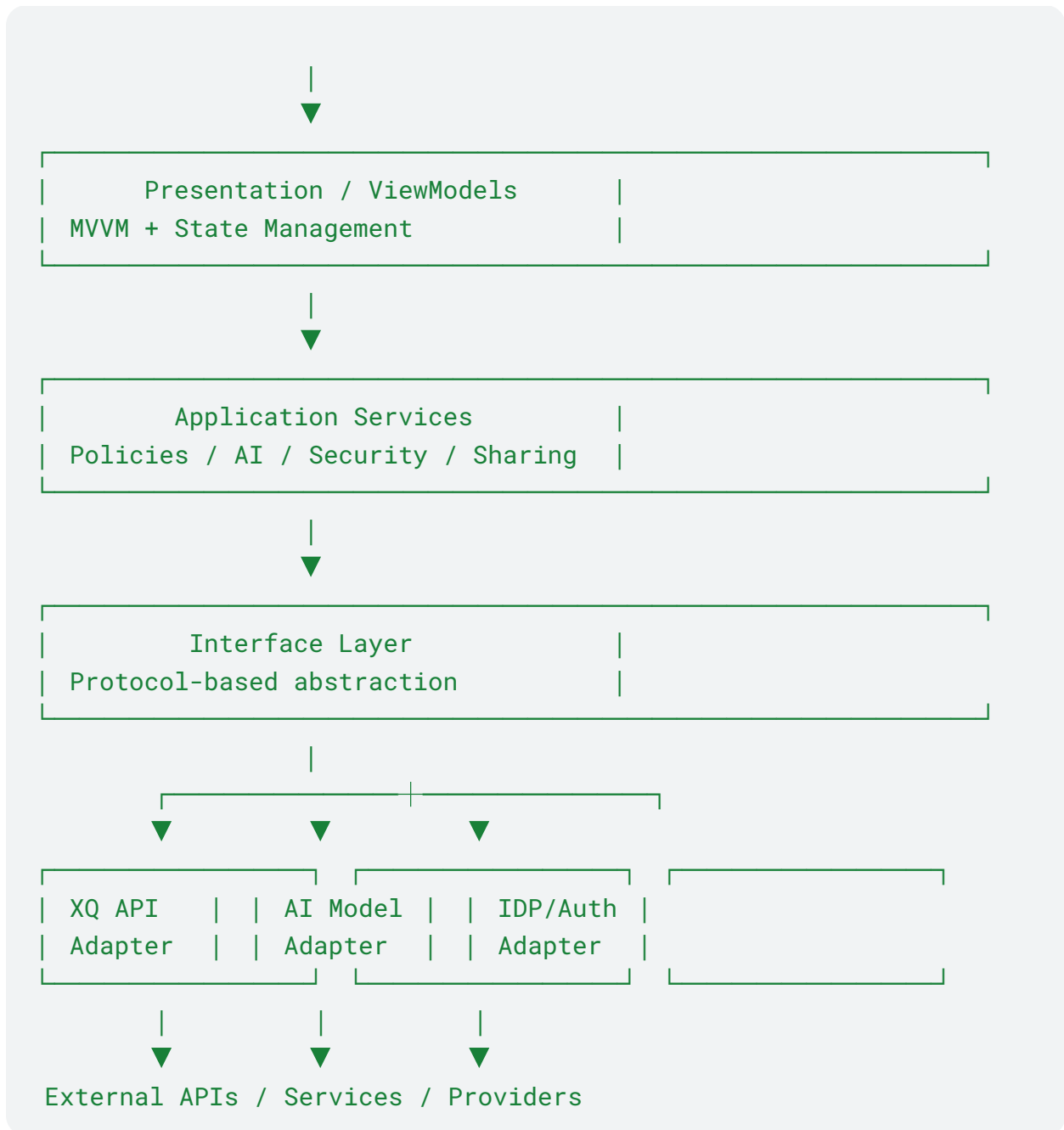
2. Core Architectural Principles

Principle	Description
Interface-Driven Architecture	All services abstracted behind interfaces
No Direct API Coupling	No business logic directly calls external APIs
AI Provider Agnostic	Interchangeable AI models/providers
Zero Trust Data-Centric Security	Protection follows data
Offline-First	Core functions operate without connectivity
Policy-Driven Runtime	Runtime behavior dynamically controlled
Cross-Platform Core	Shared architecture for iOS/Android/Desktop
Localization-Native	All UI text externally managed
Adaptive UI	Layout auto-adjusts for localization/accessibility
Enterprise Extensible	Modular enterprise integrations

3. High-Level Architecture

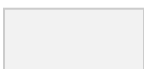
None

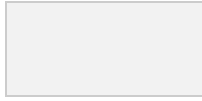




4. Mobile Platform Stack

iOS Technologies





Layer	Technology
UI	SwiftUI
State	Combine / Observable
Architecture	MVVM-C
Dependency Injection	Resolver / Factory
Persistence	CoreData / SQLite
Encryption	Secure Enclave + CryptoKit
Background Tasks	BGTaskScheduler
Networking	URLSession
Offline Sync	Custom sync engine
Logging	Unified logging framework
Animations	SwiftUI + Core Animation
AI Runtime	CoreML / ONNX Runtime
File Viewer	QuickLook + Custom Renderers

5. Interface-Based Architecture

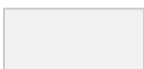
Requirement

No direct API calls allowed anywhere in business logic.

5.1 Interface Standards

All integrations must use interfaces.

Example:





Swift

```
protocol RepositoryProvider {  
    func authenticate()  
    func listFiles(path: String)  
    func downloadFile(id: String)  
    func uploadFile(file: SecureFile)  
}
```

Implementations:

- SharePointProvider
- GoogleDriveProvider
- SMBProvider
- LocalVaultProvider
- XQ API
- OAUTH

Business logic never references provider implementations directly.

5.2 Dependency Injection

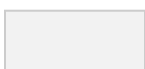
All providers resolved dynamically.

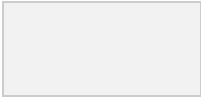
Swift

```
class RepositoryService {  
    init(provider: RepositoryProvider)  
}
```

Benefits:

- Testability
- Mock support
- Runtime provider swapping
- Version compatibility
- Enterprise customization





6. XQ API Abstraction Layer

Requirement

Support multiple XQ API versions simultaneously.

6.1 XQ API Interface

Swift

```
protocol XQSecureAPI {  
    func encrypt()  
    func decrypt()  
    func applyPolicy()  
    func revokeAccess()  
    func validateAccess()  
}
```

XQ API - <https://xq.stoplight.io/>

XQ SDK - <https://github.com/XQ-Message-Inc/>

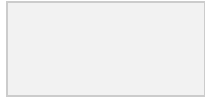
6.2 Version Adapters

None

```
XQAPIv1Adapter  
XQAPIv2Adapter  
XQAPIv3Adapter
```

Version resolution:

- Dynamic capability negotiation
 - Backward compatibility support
 - Feature flag mapping
-



6.3 API Gateway Layer

All network traffic routed through:

- Request normalization
 - Retry orchestration
 - Token refresh
 - Telemetry
 - Audit logging
 - Security validation
-

7. AI Architecture

7.1 AI Provider Abstraction

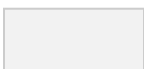
Swift

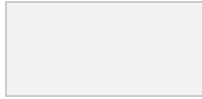
```
protocol AIProvider {  
    func classify(document: Data)  
    func generateSummary(text: String)  
    func evaluateRisk(context: RiskContext)  
}
```

Supported Providers:

- OpenAI
 - Anthropic
 - AWS Bedrock
 - Azure OpenAI
 - Local CoreML Models
 - ONNX Models
 - Enterprise custom models
-

7.2 AI Model Switching





Runtime configurable:

- Per tenant
- Per policy
- Per data classification
- Offline fallback

Example:

- Local model for CUI
- Cloud model for summarization

7.3 AI Governance Engine

Functions:

- Classification
- Sensitivity scoring
- Risk assessment
- Policy recommendations
- Behavioral analysis
- Sharing validation

8. Fuzzy Logic Policy Engine

Purpose

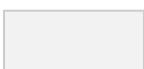
Policies are adaptive rather than binary.

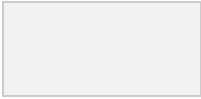
Instead of:

None

ALLOW / DENY

System evaluates:



- 
- Risk score
 - Device trust
 - User behavior
 - Location
 - Sensitivity
 - Sharing patterns
 - Time
 - Session anomalies
-

8.1 Example

None

IF:

External User

AND Sensitivity = High

AND Device Trust = Medium

THEN:

Allow View Only

Disable Download

Require MFA Every 5 Minutes

Enable Heavy Watermark

8.2 Policy Engine Architecture

None

Policy Inputs



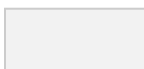
Risk Evaluation Engine

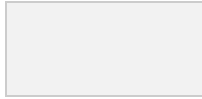


Fuzzy Logic Processor



Runtime Controls





XQ Enforcement Layer

9. Identity & Group Model

9.1 Identity Providers

Supported:

- Microsoft Entra ID
- Okta
- Google Workspace
- Ping
- AWS IAM Identity Center

9.2 Gmail Group Logic

Requirement:

Gmail identity acts as group container.

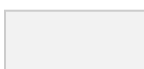
Example:

None

`project-team@gmail.com`

Automatically becomes:

- Shared secure workspace
- Shared folder boundary
- Shared governance container





9.3 Dynamic Group Policies

Groups inherit:

- Labels
- Access policies
- Sharing restrictions
- Retention rules

10. Shared Folder Collaboration Model

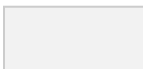
Invite-Based Workspace Creation

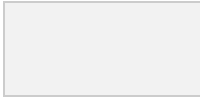
User Flow:

1. User invites participants
2. App creates encrypted workspace container
3. Policies applied automatically
4. Shared folder established
5. Dynamic membership maintained

10.1 Shared Workspace Characteristics

Capability	Description
Persistent encryption	Data remains protected
Membership governance	Dynamic access
Revocation	Immediate removal
Activity logging	Full auditability
Offline sync	Controlled secure cache





11. Localization & Language System

11.1 Multi-Language Architecture

Requirement:

All UI text externalized.

11.2 Localization Files

None

```
/en/messages.json  
/fr/messages.json  
/de/messages.json  
/jp/messages.json
```

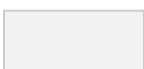
Example:

JSON

```
{  
  "home.title": "Home",  
  "files.import": "Import File",  
  "risk.warning": "Sensitive file detected"  
}
```

11.3 Runtime Language Switching

Features:





- Dynamic reload
 - No app restart required
 - Tenant-specific language packs
 - OTA translation updates
-

11.4 Adaptive UI

Requirements:

- Auto-resizing buttons
 - Flexible containers
 - RTL support
 - Dynamic type scaling
 - Accessibility-first layouts
-

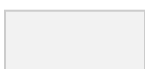
12. Animation System

Requirement

Every UX element must animate.

12.1 Animation Categories

Type	Requirement
Entry	Fade + motion
Exit	Scale + fade
Click	Spring response
Navigation	Smooth transitions





Risk Alerts Contextual emphasis

File Actions Gesture animation

12.2 Animation Framework

Use:

- SwiftUI animations
- Core Animation
- Motion interpolation engine
 - Research the best UI framework for centralized Design control. Prioritize a cross platform module.

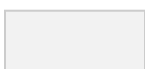
Performance requirements:

- 60 FPS minimum
- Reduced motion accessibility support

12.3 UI Guidance

For a modern iOS-first app that also needs strong Android consistency, the top choices in 2026 are:

Framework / Module Stack	Native Feel	Cross-Platform Consistency	Performance	Enterprise Suitability	Best Use Case
Kotlin Multiplatform + SwiftUI + Jetpack Compose	Excellent	High (logic shared)	Native	Excellent	Enterprise, security, AI, regulated apps





Flutter	Very Good	Excellent	Near-native	Very Good	Pixel-perfect UX/UI consistency
React Native	Good	Good	Near-native	Good	Fast iteration, JS teams
Compose Multiplatform	Very Good	Excellent	Near-native	Emerging	Kotlin-centric shared UI

Based on your requirements from prior discussions — enterprise security, modular architecture, interchangeable AI models, policy enforcement, adaptive UI, multilingual support, iPad + iPhone support, and likely long lifecycle maintenance — the strongest architecture is:

Recommended Architecture

1. Core UX/UI Stack

iOS

- SwiftUI

Android

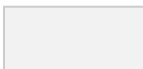
- Jetpack Compose

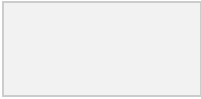
Shared Layer

- Kotlin Multiplatform

This gives:

- True native UX on each platform
- Shared business logic
- Shared networking
- Shared policy engine



- 
- Shared security layer
 - Shared AI orchestration
 - Shared localization framework
 - Shared design tokens

This is increasingly considered the enterprise-grade approach for regulated/security-sensitive mobile applications.

Why This Is Better Than Flutter

Flutter is excellent for UI consistency, but your app requirements lean heavily toward:

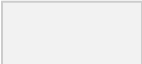
- security
- native OS integrations
- policy enforcement
- background processing
- device posture
- identity providers
- external SDK integrations
- secure storage
- MDM compatibility
- AI model abstraction
- enterprise compliance

These are areas where native platform access matters.

Kotlin Multiplatform lets you:

- share 60–80% of code
- maintain fully native UI
- avoid bridge performance issues
- integrate deeply with iOS and Android security APIs

This is especially important for:

- Zero Trust enforcement
 - secure enclaves
 - device attestation
 - offline policy evaluation
 - AI model switching
 - encrypted local storage
- 



Best UX/UI Modules To Standardize

Design System Layer

Use:

- Material 3 tokens internally
- Apple Human Interface Guidelines on iOS
- Shared design tokens via KMP

Recommended structure:

None

Shared Design Tokens

├── colors

├── typography

├── spacing

├── iconography

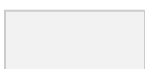
├── motion

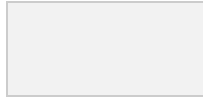
├── accessibility

└── responsive breakpoints

Best Native UI Modules

Navigation





iOS

- UINavigationController
- SplitView for iPad

Android

- Navigation Compose

State Management

Recommended:

- MVVM + unidirectional data flow

Use:

- StateFlow
- Combine
- Shared ViewModels via KMP

Best Cross-Platform UI Component Systems

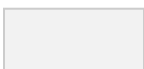
Best Overall

Compose Multiplatform

Best future-facing option if you later want:

- desktop
- web
- Android Automotive
- embedded AI terminals

But:





- iOS support is still maturing

Best Stable Enterprise Choice

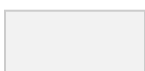
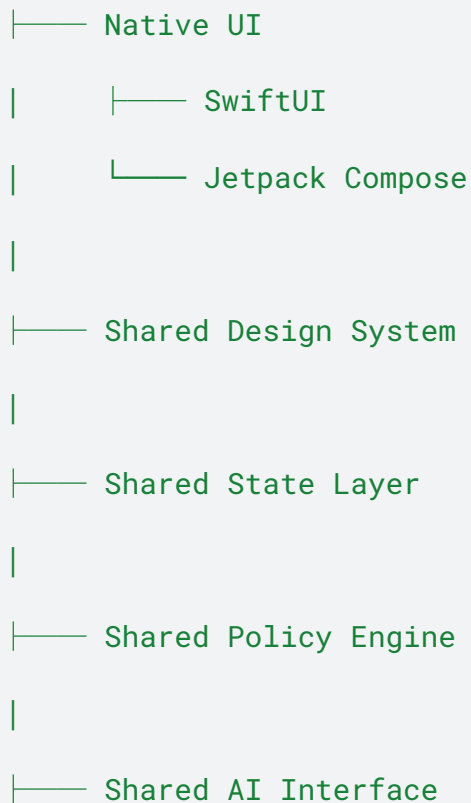
SwiftUI + Compose + KMP

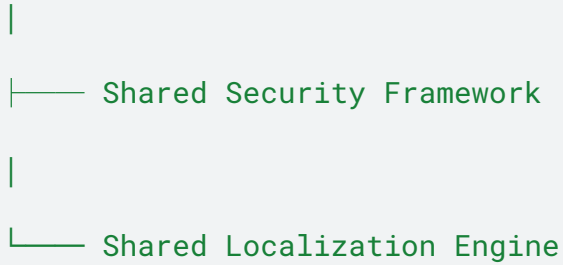
This is currently the safest long-term architecture.

Recommended Modular UX/UI Architecture

None

Presentation Layer





Important UX/UI Requirements For Your App

Given your previous requirements, your UI framework should support:

Dynamic Policy-Controlled UI

Example:

- Hide features by RBAC/ABAC
- Dynamic labels
- Geofenced UI
- Tenant-based layouts

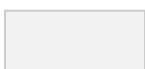
KMP handles this very well.

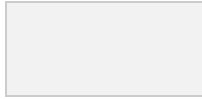
Externalized Localization

Use:

```
JSON
/localization/en.json

/localization/fr.json
```





```
/localization/ar.json
```

Do NOT hardcode strings.

Adaptive Layout System

Must support:

- iPhone
- iPad
- foldables
- Android tablets
- external monitors

SwiftUI + Compose are best-in-class here.

Accessibility

Need:

- Dynamic text
- VoiceOver/TalkBack
- keyboard navigation
- high contrast
- RTL layouts

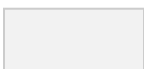
Native UI frameworks handle accessibility significantly better than Flutter or React Native.

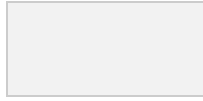
Final Recommendation

Best Overall Architecture

Kotlin Multiplatform

-





SwiftUI

-

Jetpack Compose

This gives you:

- native-grade UX
- shared business/security logic
- Android/iOS consistency
- enterprise scalability
- best long-term maintainability
- strong AI integration flexibility
- best security posture

Flutter is the second-best choice if:

- maximum visual consistency matters more than native fidelity.

React Native is best only if:

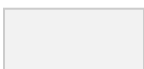
- your organization is heavily JavaScript/TypeScript-centric.

13. Resource Control System

Purpose

Prevent excessive:

- Battery drain
- AI processing
- Memory consumption
- Network utilization





13.1 Adaptive Resource Policies

Dynamic controls:

- AI throttling
- Sync batching
- Background limits
- GPU usage caps
- Cache pruning

13.2 Device-Aware Scaling

Example:

Older devices:

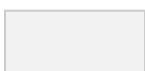
- Reduced animation complexity
- Smaller AI models
- Limited background tasks

14. Secure File System

14.1 File Protection Model

All imported files:

- Encrypted immediately
- Bound to XQ policy metadata
- Stored in secure sandbox
- Never exposed directly to iOS filesystem





14.2 Runtime Controls

Protected Viewer:

- Screenshot blocking
- Screen recording detection
- Clipboard prevention
- Background blur
- Watermark overlays

14.3 Offline Storage

Offline files:

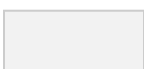
- Time-limited
- Policy-bound
- Auto-expiring
- Remote revocable

15. Integration Framework

15.1 Unified Integration Interface

Swift

```
protocol ExternalIntegration {  
    func authenticate()  
    func sync()  
    func query()  
    func upload()  
}
```





15.2 Supported Integrations

Microsoft

- SharePoint
- OneDrive
- Teams
- Exchange
- Entra ID

Google

- Gmail
- Google Drive
- Google Workspace

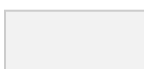
Additional

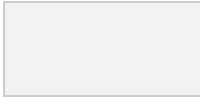
- Box
- Dropbox
- SMB
- AWS S3
- Local storage

16. Auto Update / Force Update System

16.1 Update Modes

Mode	Description
------	-------------





Silent Update	Background content/config updates
Soft Update	Recommended app upgrade
Force Update	Mandatory security update

16.2 Dynamic Configuration

Remote config controls:

- Feature flags
- Policy updates
- AI model rollout
- Kill switches
- Version gating

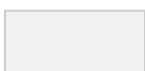
17. Cross-Platform Strategy

17.1 Shared Architecture

Core shared concepts:

- APIs
- Policy engine
- AI abstraction
- Sync logic
- Security model

17.2 Future Platform Support





Platform	Strategy
Android	Kotlin equivalent architecture
macOS	Native Catalyst
Windows	MAUI/Native
Web	Secure browser workspace

18. Security Architecture

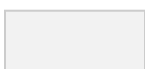
18.1 Core Security Controls

Control	Description
Secure Enclave	Key storage
Certificate Pinning	API security
Runtime integrity	Tamper detection
Jailbreak detection	Compromise prevention
Device posture	Trust scoring
Secure cache	Encrypted storage

18.2 Zero Trust Runtime

Continuous evaluation:

- Identity
- Device
- Session





- Location
- Behavior
- Classification
- Network posture
- version

19. Governance Architecture

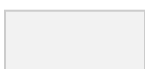
19.1 Policy Types

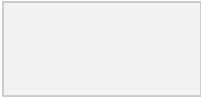
Policy	Description
Classification	Data sensitivity
Sharing	External access
Retention	Expiration
Device	Trust requirements
Geographic	Region restrictions
Runtime	Viewer protections

19.2 Label Architecture

Labels support:

- Follow MIPS
- Hierarchical inheritance
- Multi-label assignment
- AI confidence scoring
- Dynamic reclassification





20. Offline-First Sync Architecture

Principles

- Local-first operations
- Eventual consistency
- Conflict resolution
- Secure sync queue

20.1 Sync Engine

Required for local groups

Capabilities:

- Delta sync
- Retry queue
- Background sync
- Policy-aware sync
- Selective offline availability



21. Enterprise Admin Framework

Capabilities

Governance

- Policy authoring
- AI threshold tuning
- Group administration
- Access management

Audit

- File lineage
- User behavior
- Sharing events
- Device trust history

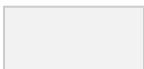
22. Observability & Telemetry

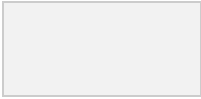
Telemetry Categories

Category	Description
Security	Threat events
UX	User interactions
AI	Model performance
Sync	Repository health
Runtime	Device performance

All telemetry:

- Policy governed
- Privacy aware



- 
- Tenant configurable
-

23. Recommended Repository Structure

None

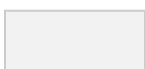
App/

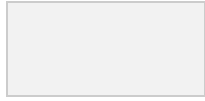
```
|— Core/
|— UI/
|— ViewModels/
|— Services/
|— Interfaces/
|— Providers/
|— AI/
|— Security/
|— Policies/
|— Localization/
|— Animations/
|— Sync/
|— Integrations/
|— Storage/
|— Utilities/
```

24. Recommended Development Phases

Phase 1

- Authentication
- File browser
- Secure viewer
- XQ encryption
- SharePoint integration
- Offline cache





Phase 2

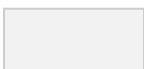
- AI classification
- Dynamic policy engine
- Secure sharing
- Gmail group workspaces

Phase 3

- Advanced governance
- AI copilots
- Multi-model orchestration
- Cross-platform sync

Phase 4

- Autonomous governance agents
- AI collaboration
- Enterprise analytics
- Secure workflow automation



Design guidance

Best Overall Native iOS Design

Things 3

- Gold standard for native iOS interaction design
- Exceptional typography, spacing, motion, and gesture behavior
- Uses Apple Human Interface Guidelines extremely well
- Great example of progressive disclosure and low cognitive load

Craft

- Beautiful editor and navigation system
- Excellent iPad + iPhone adaptive layouts
- Strong use of glassmorphism and modern Apple visual language
- Very polished animations and transitions

Fantastical

- Outstanding information density without clutter
- Excellent gesture systems and calendar UX
- Strong widget and Apple ecosystem integration

Linear

- One of the cleanest modern productivity UIs
 - High-performance feel with subtle motion design
 - Strong cross-platform consistency between iOS, macOS, and web
-

Best Apps for Interaction Design

Flighty

- Best-in-class live activity and Dynamic Island implementation
- Superb real-time UX patterns
- Excellent animation timing and visual hierarchy

Gentler Streak

- Exceptional health data visualization
- Friendly, emotionally intelligent UX
- Strong accessibility and Apple Watch integration

Dark Noise

- Minimalist interface with excellent tactile interactions
- Great use of haptics and iOS-native controls
- Highly refined microinteractions